R.B.

Dalhousie University

Psychology and Neuroscience

29th Annual
In-House Convention

1-2 May 2003

Photos far too dark (Morely)

Welcome to the 2003 Psychology and Neuroscience In-House Convention

Thanks for attending the 29th Annual In-House Convention.

Thanks should also be extended to the In-House Convention Committee for planning and organization. The committee consists of the following individuals: John Connolly, Sarah Boudreau, Nancy Garon, Troy Harker, Kris Peace, Amanda Wintink and Kristi Wright.

Friendly Reminders

*Please bring your own mug for coffee or tea!

*All presentations are listed in the name of the presenter from the abstracts submitted.

*There will be pizza and beer at the end of Friday in the 2nd Floor Lounge.

Late Additions..........

Talk title change:

T6) Sticks and stones may break my bones, but names will make me hurt me.

Stephen Lewis

New talk:

May 2, 2003 at 10:15 am.

T19) Music, Mood and Booze: Activation of implicit alcohol cognition in coping and enhancement motivated drinkers.

Sherry Stewart, Cheryl Birch, Shondalee Eisnor, & Jennifer Theakston

Cooper's (1994) model of drinking motivations distinguishes between two groups of drinkers at risk of drinking problems – those who drink to enhance positive mood and those who drink to cope with negative mood. We investigated whether these groups differ in their activation of implicit alcohol cognition following mood induction. Participants were 86 undergraduates with elevated scores on the Enhancement or Coping Motives (EM or CM) subscale of the Drinking Motives Questionnaire. Participants in each motive group were randomly assigned to a musically-induced positive or negative mood condition. A modified Stroop (1935) task, involving color-naming of alcohol and non-alcohol control words, was administered following mood induction. As hypothesized, EM drinkers in the positive (but not negative) mood condition showed significantly longer color-naming speeds for alcohol vs. control words indicating interference in color-naming by the content of the alcohol words. Even though CM drinkers in the positive mood condition did not show evidence of selective processing of alcohol cues (as expected), CM drinkers in the negative mood condition also failed to show evidence of alcohol interference. Overall, CM drinkers showed significantly longer color-naming latencies than EM drinkers suggesting possible ceiling effects in the CM group. Theoretical and clinical implications will be discussed.

New poster

P11) **How do different mouse strains differ on measures of locomotion, exploration, anxiety, learning and memory?**

Dirk Luchtman and Richard Brown

Transgenic mouse models of human diseases provide an animal model for understanding the underlying mechanisms of these diseases. It is hoped that, eventually, these models will translate into new treatments of human (neurological) diseases. Researchers are assembling test batteries in order to compare behavioral expressions of transgenic and mutant mice. The data that is produced in these experiments can be used to build up electronic databases, like the Mouse Phenome Database of the Jackson Laboratory.

As part of the Phenome Project, we are investigating the behavioral differences in 5 strains of mice: SPRET/Ei, MOLF/Ei, C3H/HeJ, SJL/J and A/J. We are testing these mice for anxiety-like behaviors on the elevated plus maze, and light-dark box; locomotion and exploration in the open field; motor learning on the rotarod; and spatial learning in the Barnes maze and Morris water maze. We are also testing these mice for olfactory discrimination learning and visual acuity. Our poster compares the 5 strains of mice on these behaviors and examines the influence of body weight, anxiety, exploratory behavior and visual acuity on performance in tests of learning en memory.

Thursday, May1, 2003
Room 4258, Life Sciences Centre

9:15 Introduction and brief history of convention (Vin LoLordo)

9:30-10:30 Session 1.

T1) Jason Ivanoff
T2) Troy Harker
T3) Elizabeth Service
T4) Kristie Dukewich

10:30-10:45 Coffee Break

10:45-12:00 Session 2.

T5) John Barresi
T6) Stephen Lewis
T7) Sandra Reyno
T8) Kimberly Good
T9) — —

12:00-2:00 Lunch Break and Posters (Psychology 2nd Floor Lounge)

P1) Lisa Currie
P2) Beverly Butler
P3) Slobodan Mirkovic
P4) Crystal Lowe-Pearce
P5) Lianne Stanford
P9) Mohammad Inanlou

2:00-3:30 Session 3.

T10) Eric Morris
T11) Amanda Wintink
T12) Rebecca Douglas
T13) Kathleen Hourihan
T14) Tracy Taylor
T15) Megan Therrien

Friday, May 2, 2003
Room 4258, Life Sciences Centre

9:30-10:30	Session 1.
	T16) Samuel Deurveilher T17) Amanda Poulin T18) Marie-Josee Lefaivre T19) — —
10:30-10:45	Coffee Break
10:45-12:00	Session 2.
	T20) Kerry Walker T21) Aimee Wong T22) Jamesie Coolican T23) Donald Mitchell T24) Dennis Phillips
12:00-2:00	Lunch Break and Posters (Psychology 2nd Floor Lounge)
	P6) Amy MacPherson P7) David Westwood P8) Jeannette McGlone P10) Martin Willamson
2:00-3:30	Session 3.
	T25) Tamara Franklin T26) Meghan Bain T27) Sarah Johnson T28) Denise Bernier T29) Karthika Devarajan T30) Steve Shaw
4:00	Pizza and Beer in the 2nd Floor Lounge

T1) You snooze - sometimes you win, sometimes you lose.

Jason Ivanoff & Benjamin Rusak

The process of encoding a recently formed memory trace into a long-lasting form is called consolidation. Overnight sleep is believed to facilitate the consolidation process. We tried to determine whether a 2-hr afternoon nap helps to consolidate newly formed memories in a way similar to what has been claimed recently for overnight sleep. Results of a pilot study tentatively suggest that an afternoon nap can help consolidate some types of memories, but hurt others. Moreover, the results also suggest nap-associated improvements in tasks requiring inhibitory executive functioning.

T2) Neuropsychological Assessment of Memory using Event-Related Potentials

Troy Harker and John Connolly

The use of cognitive event-related potential recordings (ERP) in conjunction with standardized neuropsychological testing instruments provides a useful, non-invasive clinical tool for investigating and measuring the cognitive abilities of non-communicating clinical populations. In light of previous success using this approach with respect to speech and language processing, the goal of the present study was to determine if neuropsychological tests of higher-order cognitive functions such as learning and memory could also be successfully adapted for ERP presentation. Thirty undergraduate students (15 male, 15 female) will be tested on both the regular and the ERP adapted version of one to two standardized neuropsychological tests. ERP results will be examined to identify possible characteristics that can differentiate between correct and incorrect responses. Performance on the ERP version will also be cross-validated with performance on the standardized version of the test to determine concurrent validity. Results will be discussed in terms of both the potential clinical utility of the ERP adapted tasks as well as the important insight into the neurophysiological mechanisms, by which individuals accomplish such memory tasks, afforded by the ERP component patterns.

T3) Phoneme-level information in short-term memory for wordlike items

Elisabet Service

Verbal short-term memory has in the majority of studies been investigated as the capacity for immediate ordered reproduction of a series of words, letters or digits. In three studies, we explored memory and forgetting of sub-lexical units, for instance syllables and single phonemes, in pseudowords. Our studies suggest that immediate recall is affected by information representations on different layers of phonological complexity (whole words, as well as syllables and single vowels and consonants). Poor phonological memory, linked to developmental dyslexia and language learning difficulties, can result from poor encoding or storage of traces on these sub-lexical levels, leading to unsuccessful retrieval of whole words at recall. Poor representations on the sublexical level can also be revealed in ERP studies of tasks relying on precise phonological representations.

T4) **Limiting the number of inspections during a difficult search task: Implications for the seriality of search**

Kristie Dukewich & Dr. Ray Klein

Treisman and Gelade (1980) changed the literature on visual search when they made the distinction between feature search and conjunction search. Serial search is inferred from a linear increase in reaction times as set size increases. Triesman and Gelade's Feature Integration Theory (FIT) accounts for this by assuming that attention is required to glue together the features of a conjunction target, making serial inspection necessary to find the target. Recent challenges to this view suggest that conjunction search uses both serial and parallel processes to localize the target. If serial self-terminating search is responsible for the linear functions found in conjunction search tasks, then accuracy on a task that limits display duration should be predictable based on search rates estimated from slopes obtained from a typical, long display duration task. We found that people are not limited by typical conjunction search rates, and are able to extract more information than predicted by the serial search model. Our results converge with other challenges to a strictly serial self-terminating inspection process, and point to the likelihood of a hybrid model wherein serial and parallel mechanisms operate in tandem during difficult searches.

T5) **From "Nobody Nowhere" to "Somebody Somewhere": A case study in the phenomenology of autism and of the discovery of the interpersonal world.**

John Barresi

In three astonishing autobiographies, Donna Williams describes a phenomenological journey from her own private world of childhood autism to her adult discovery of - and eventual entry into - the interpersonal world. In this talk, I will analyze these autobiographies with a focus on the transformations in Williams' experience and understanding of self and other and of the role that her body plays in her discovery of and entry into the interpersonal world. In order to enter this world that unifies self and other, she has to overcome the alternating perspectives of the first- and third-person worlds of her earlier experiences, where Donna-for-others differed from Donna-for-self. Since the battle ground of this opposition between the two worlds is her own body, a major part of her transformation involves a process by which Donna comes to experience her self as a unified person who is both owner & agent of her body. Only once she has done this can she enter freely into an interpersonal world with other embodied agents and persons.

T6) Elucidating the link between bullying, mood and self-injury among young people

Stephen Lewis

Self-injurious behaviour (SIB) can be best understood as intentionally harming oneself (e.g. cutting) in order to provide rapid, albeit ephemeral, relief from psychological turmoil (Briere & Gil, 1998; Favazza, 1996, 1998). Typically, SIB onsets during adolescence and persists well into adulthood if left untreated. The plight of SIB carries with it many risks, including serious injury, and accidental or intentional suicide, all of which accentuate the psychological pain of the self-injurer (Favazza, 1998). Though precise aetiologies for SIB remain elusive, several correlates have been identified. For instance, it is well accepted that abuse associates with SIB (Lipshitz et al., 1999). It is also well accepted that abuse associates with mood problems (Andrews, 1995; Browne & Finkelhor, 1986). Taken together, it is interesting to speculate how abuse and mood are associated with SIB. Specifically, it is interesting to consider bullying as a form of abuse due to its ubiquity amongst young people (Bond et al., 2001; Nansel et al., 2001) and its association with mood disturbances (Bond et al., 2001). Despite this relationship, little work has been done to investigate bullying and SIB. The dearth of work completed suggests that bullying may be a correlate of SIB (Bhugra et al., 2002; Davies & Cunningham, 1999). In sum, little is known about the link between bullying, mood and SIB. As a result, the present study aims to elucidate this link by regressing these variables in order to construct a mediational model. Findings, clinical implications and directions for future research will be discussed.

T7) The influence of Parental Factors on Recurrent Headache in Adolescents

Sandra Reyno

Functional headache or headache not due to an identified organic cause is one of the most common somatic complaints in adolescents. Recurrent functional headache in adolescents is associated with a number of factors. These include child mental health factors (i.e., depression, anxiety), child demographic factors (i.e., gender, age), and familial factors (i.e., socioeconomic status, family headache history). Parental factors associated with recurrent functional headache in adolescents include parental encouragement of child illness behavior and parental modeling of illness behavior. It is not known how these parental factors impact on headache-related impairment (pain intensity), functional disability (restriction in performing social, academic, and physical tasks), or handicap (inability to fulfill social, academic, or physical roles). In the proposed research, an Internet-based study will be used to evaluate a model of headache in adolescents. The model will examine hypothesized relationships between parental factors and headache-related impairment, functional disability, and handicap in adolescents.

T8) **Functional Magnetic Resonance Imaging of the Olfactory System in patients with First Episode Psychosis: Preliminary Data from a Normosmic Subgroup**

Kimberley P. Good, Heather Schellinck, Larry Gates, Richard Brown, Sarah Boudreau, Sheldon Choo, E. Mark Haacke, Lili Kopala

Classical neuroanatomic techniques and lesion studies have well delinated the neural substrates involved in olfactory processing. Functional imaging has extended these findings by allowing the identification of brain regions engaged by different cognitive-olfactory tasks. The purpose of this ongoing study is to use FMRI to examine the functional topography of the olfactory system in patients with first episode psychosis and healthy control subjects. Fourteen normosmic subjects participated: 8 healthy controls and 6 patients with DSM-IV confirmed diagnosis of schizophrenia. A manually-controlled olfactometer delivered the stimulus (amyl acetate) to the subjects during the FMRI scan. Twenty six second on-off epochs (5 odour, 4 no-odour) were presented.

As predicted, healthy control subjects demonstrated activity in previously defined olfactory regions (left superior, middle and inferior frontal gyri, right orbitofrontal cortex, and right inusla). Activation in left basal ganglia and left superior temporal gyrus was also observed. Patients demonstrated activity in the left superior frontal gyrus and posterior cingulate.

In this preliminary dataset, secondary olfactory cortices did not show the predicted activity in the patient sample in response to the presented odour. As patients with psychosis have been documented with olfactory identification deficits (OID), these data broaden our understanding of the functional brain abnormalities in these patients. Further research by our lab will examine cognitive-olfactory tasks in psychosis patients with confirmed OID.

T9) TBA

T10) **Oops, We Gave You the Wrong List: List-Method Directed Forgetting using Threat and Anxiety Related Words with Those High and Low in Social Anxiety**

Eric Morris

Directed forgetting is the conscious effort to forget information. Studies in this area typically use one of two methods to elicit the directed forgetting effect: the list method and the item method. The list method was used in the present study and involved presenting a list of neutral, positive, threatening and anxiety producing words and telling participants to either remember or forget the preceding list, followed by a free recall test. The first stage of the present study involves running normal controls in order to confirm that the program we created to produce the directed forgetting effect is effective. The talk will focus on this stage of the study. The second phase of the study will involve administering the program to high and low socially anxious individuals. The goals of the study are two-fold: To create a program that consistently produces the directed forgetting effect and to determine whether high or low socially anxious individuals have more or less trouble actively forgetting threat and anxiety related words.

T11) Jack and Jill swam in a pool to find a hidden platform. Jack swam by, like the other guy, but Jill found it right after

Amanda Wintink

Several lines of research suggest that anxiety behavior in amygdala-kindled rats is related to changes in the hippocampus, such as increased serotonin and GABA receptors, decreased NMDA and glucocorticoid receptors, and decreased cFOS expression. Behaviorally, amgydala-kindled rats primarily express anxiety behavior in novel environments, which further suggests a hippocampal involvement. The present experiment was designed to test whether amygdala-kindled rats were impaired on a hippocampal-dependent task, as a behavioral assessment of hippocampal functioning. For this experiment, adult male and female rats were trained to find a hidden platform in the Morris watermaze using a delayed-matching-to-sample (DMTS) paradigm with a 60-minute delay between the sample (i.e., trial 1) and matching (i.e, trial 2) trials. Rats were trained on the task for 5 days prior to kindling, in which only the platform location changed daily. Rats were then kindled and tested on the task once a week during kindling, for a total of 6 weeks, with a new platform location each week. The results showed that kindling had a negative impact on watermaze performance primarily for male rats.

T12) Memory for Details of Visual Scenes: The Role of Psychopathic Traits and Emotional Valence

Rebecca Douglas

Emotion has been shown to play a complex role in memory. It appears that with emotional stimuli, central details are retained better than peripheral details, and that central details of emotional events may be retained better than those of non-emotional events. In contrast, individuals with psychopathic traits display virtually the same memory for central and peripheral details of emotional stimuli, and show no difference between memory patterns for emotional and neutral stimuli. However, as the majority of research with both individuals high in psychopathic traits and non-psychopathic populations has focused on negative emotional events, the degree to which these findings apply to positive stimuli is unknown. Thus, the objective of this study is to extend previous research findings concerning the influence of emotion on memory to positive stimuli. Also to be examined is the relationship that psychopathic traits may have with stimuli of negative, neutral and positive emotional content. Ninety participants will be assessed on their degree of psychopathic traits using the self-report Psychopathic Personality Inventory (Lilienfield & Andrews, 1996). A positive, negative and neutral scene, selected from the International Affective Picture System (Lang, Bradley & Cuthbert, 1999), will be presented to each participant. Participants will then be tested on their memory for central and peripheral details of each scene. The relationship between degree of psychopathic traits, emotional valence, and memory for central versus peripheral details will be examined. It is hypothesized that previous findings will be replicated, and that patterns demonstrated with negative stimuli may extend to positive stimuli.

T13) **Does rehearsal time affect the way in which information is forgotten?**

Kathleen Hourihan

The present experiment examined the effects of rehearsal time in an item method directed forgetting paradigm. Participants were presented with a list of words and were instructed to recall the words unless they received a forget cue, which occurred on half of the trials, at varying delays after word presentation. After studying the words, participants completed a recall test of all words (regardless of initial cue), a recognition test, and a second recall test. Results showed that while a directed forgetting effect was obtained for all three tests, the magnitude of the effect was reduced for the recognition and second recall test, and that words that received a forget cue at a longer delay were more frequently recalled than words that received the short delay. Results were discussed in terms of retrieval inhibition and selective rehearsal.

T14) **The role of spatial attention in directed forgetting: Take 1**

Tracy Taylor

This study determined whether *directed forgetting* of peripherally presented words involves processes related to visuo-spatial attention. This was tested by examining the magnitude of inhibition of return at the location of to-be-remembered and to-be-forgotten words. Using a modified item method paradigm, words appeared one at a time to the left or right, followed by a visual *Remember* or *Forget* instruction at center. A target dot was then presented either in the same peripheral location as the preceding word or in a different location; participants made a speeded response to localize this target. Despite the fact that a *directed forgetting* effect occurred for both recognition and recall, there was no effect of memory instruction on the magnitude of inhibition of return (as measured by the reaction times to the subsequent targets). This suggests that information about the to-be-remembered and to-be-forgotten words is extracted independently of the spatial information that marks the presentation episode.

T15) **The role of spatial attention in directed forgetting: Take 2**

Megan Therrien & Tracy Taylor

This is a follow-up to a study by Taylor that indicated that directed forgetting of a to-be-forgotten word does not involve inhibition of that word's location. The present study attempted to replicate this result, using tones rather than visual stimuli to designate words as to-be-forgotten or to-be-remembered. Words appeared to the left or right, followed by a tone instructing participants to remember the word or to forget it. Participants then made a speeded response indicating the location of a target dot that appeared subsequently to the left or right. The magnitude of inhibition of return was evaluated at the location of both to-be-forgotten and to-be-remembered words. (At present, the results are unknown).

T16) How coffee wakes up your brain: caffeine-induced activation of Arousal-Related Aminergic neurons in rat

Samuel Deurveilher, Jeremy Murphy, Joan Burns and Kazue Semba.

Caffeine is the most widely used psychostimulant in the world. However, neuronal mechanisms mediating its stimulatory effect are not well understood. We recently reported, using c-Fos as a marker for neuronal activation, that psychostimulant doses of caffeine induce activation of hypocretin/orexin-containing neurons which play an important role in arousal. Here we examined whether other components of the arousal system, specifically aminergic neuronal groups, are also activated by caffeine, by using dual immunostaining for c-Fos and aminergic transmitter markers. We found that low locomotor-stimulant doses of caffeine (10 and 30 mg/kg, i.p.) increase Fos immunoreactivity in noradrenergic neurons in the locus coeruleus and histaminergic neurons in the tuberomammillary nucleus, as well as in orexin-containing neurons of the hypothalamus, but not in serotonergic neurons in the dorsal raphe nucleus. A higher, locomotor-depressant dose of caffeine (75 mg/kg, i.p.) induces c-Fos expression in all of these neurons. Dopaminergic neurons in the ventral tegmental area and substantia nigra did not respond to caffeine with changes in c-Fos expression. These results indicate that caffeine induces c-Fos expression differentially among aminergic arousal-related cell groups. The noradrenergic and histaminergic neurons and orexin-containing neurons which are activated at low doses of caffeine might play a role in mediating behavioral activation induced by caffeine.

T17) Confocal Microscopic Analysis of the Colocalization of Vesicular Glutamate Transporters in Cholinergic Neurons of the Rat Basal Forebrain: Do cholinergic neurons release glutamate?

Amanda Poulin and Kazue Semba

In addition to its well known role in cognition, the basal forebrain (BF) is implicated in the regulation of behavioural state. Neurochemically, these functions have been attributed primarily to the cholinergic, and to some extent, the GABAergic neurons in this brain region. Recent evidence also suggests the involvement of a glutamatergic system within the BF. Glutamate may act as a major transmitter in the BF, and may also be co-released from a subpopulation of cholinergic BF neurons. We examined the possibility of glutamate colocalization in BF cholinergic neurons by using antibodies to vesicular acetylcholine transporter and vesicular glutamate transporters (VGluT)-1/2/3, three recently identified transporters responsible for glutamate uptake into synaptic vesicles. Confocal microscopy was used to examine coimmunofluorescence for VAChT and VGluT 1/2/3 in the cell bodies of BF cholinergic neurons and their terminals in selected target areas. In the BF, scattered neurons in the horizontal limb of the diagonal band, ventral pallidum, substantia innominata, and globus pallidus were immunoreactive (ir) for both VAChT and VGluT3; however, no cell bodies in the BF were double-labeled for VGluT1/2 and VAChT. No fibers or terminals in the motor, somatosensory, piriform and entorhinal cortices were coimmunoreactive for VGluT1/2/3 and VAChT; similarly, VAChT-ir terminals within the BF were not VGluT1/2/3-ir. The reticular nucleus of the thalamus, which receives cholinergic projections from the BF, was also devoid of coimmunofluorescence for VAChT and VGluT1/2/3. In some of the above areas with cholinergic innervation, VGluT1/2/3-ir and VAChT-ir terminals were segregated in distinct

patterns, whereas in others they were intermixed. These results suggest that VGluT3 is present in a small population of cholinergic BF neurons. The presence of VGluT3 in a subpopulation of BF cholinergic neurons suggests that terminals of some BF cholinergic neurons might release both acetylcholine and glutamate.

T18) **Mothers' Views about Stress and Depression**

Marie-josée Lefaivre

Despite the fact that depression is a prevalent mental health problem in women, very little is actually known about women's views and opinions regarding stress and maternal depression. A few qualitative studies suggest that women in rural communities tend to normalize their depressive experiences by seeing it as part of their lives as mothers and housewives. Other studies indicate that these women are more likely to present somatic symptoms associated with depression or distress. Previous research also suggests that individuals often characterize their depression symptoms as part of a physical illness as a result of stigma and that perceived stigma constitutes a barrier to seeking help and to treatment adherence. Consequently, a better understanding of mothers' perception towards depression and social stigmas could be beneficial given that these perceptions are reflected in their coping skills including help-seeking behaviours. The goal of the following research project aims at examining (1) rural and urban mothers' views about depression and its symptoms, (2) their awareness of depression indicators, (3) their causal belief about maternal depression and (4) their perception of social stigma associated with this mental health issue. Mothers (19 to 49 years of age) randomly selected from the Nova Scotia Atlee Perinatal Database will be invited to participate in this study. This research is being conducted within the Maternal Depression Project of the Family Help Program: Bringing Health Home (Pediatric Pain Lab, IWK Health Centre) and in collaboration with the Reproductive Care Program.

T19) TBA

T20) **Hear no consonant. Speak no consonant. Read no consonant.**

Kerry Walker

People show great variation in their language and reading proficiency, even within the normal population. Historically, language function was viewed as somewhat separate from basic perceptual and cognitive processes. Some experts now suggest that problems with auditory and visual perception underlie language disorders. In particular, dyslexia has been theorized to be a temporal processing disorder, in which children cannot normally perceive auditory and/or visual events that occur quickly in time. Over the past 15 years, many studies have provided empirical support for this theory of language function, but most of this research has specifically targeted highly-selected and clinically-impaired children or adults. Therefore, the present study will aim to identify the basic perceptual skills that are most highly correlated with language and reading abilities in normal subjects (ages 6 to 17). The same subjects will complete various tests of auditory and visual temporal processing, language, and reading performance. Statistical analysis will then identify the basic perceptual skills that covary with language and reading proficiency. This research will help us understand the development of language in the general population, and will have implications for the treatment of developmental language delay in children.

T21) Behavioral assessment of visual discrimination, pattern discrimination and visual acuity in seven strains of mice

Aimee A. Wong and Richard E. Brown

Previous research has demonstrated that mice are capable of brightness discrimination and visual pattern discrimination and that it is possible to quantitatively determine the visual acuity of the mouse. Based on the procedure of Glen Prusky (Prusky et al., 2000, Vision Research, 40, 2201-2209), we have developed a protocol using a computer-based, two-alternative swim task that evaluates visual discrimination, pattern discrimination and visual acuity in one apparatus. Seven strains of mice (BALB/cByJ, C3H/HeSnJ, C57BL/6J, Coloboma, DBA/2J, FVB/NJ and MOLF/Ei) were tested on 8 trials/day for 8 days on each of the three tests. There was a significant strain difference in visual ability in all three tests. Mice with normal vision (C57BL/6J and DBA/2J) performed very well in these tasks. Mice with poor vision (BALB/cByJ) took longer to learn the tasks than mice with normal vision and mice with retinal degeneration (C3H/HeSnJ, Coloboma, FVB/NJ, MOLF/Ei) performed only at chance levels. Because measures of leaning and memory on visuo-spatial tasks may be affected by visual ability, we correlated performance in the Morris Water Maze with the visual acuity of each strain. We found that visual acuity was significantly correlated ($r = -0.998$) with the latency to find the platform in the Morris Water Maze. This suggests that the visual ability of each strain is a confounding variable in visuo-spatial learning and that visual acuity, rather than spatial memory capability, may account for some of the strain differences in mice in the Morris Water Maze.

T22) Investigating Spatial Biases in Face Perception

Jamesie Coolican, BSc. and Gail Eskes, Ph.D

We were interested in the reported left-sided perceptual bias in interpreting facial emotion and in facial recognition. Our purpose was to replicate the reported bias in normal subjects and to examine whether this phenomenon was related to global/holistic face processing per se. Twenty-one undergraduate students participated in the study. In Study 1, subjects were shown chimeric faces created by combining half of a smiling face and half of a neutral face. They were asked to judge which of two faces (smile on left vs smile on right side of the face) was "happier". Results showed a strong bias for judging the face with the left-sided smile as happier. In a second condition, the chimeric faces were inverted. The left-sided bias in emotion judgement remained, suggesting this judgement was not based on global face processing. In Study 2, composite faces from the left or right side of a neutral face were created and the subjects were asked to identify which composite most nearly matched the original face. In this case, a significant (but smaller) left-sided bias for matching a composite was seen. In addition, top faces were significantly more likely to be judged as more similar to the original than bottom faces, suggesting the recognition process was related to vertical spatial position of the face as well. When the faces were inverted, vertical spatial position ceased to affect judgements, while the left-sided bias remained. Implications of these findings for neurocognitive models of face processing will be offered for discussion.

T23) How to Grow up and be normal: Lessons from the visual system and accenture. It is not the variety or amount of early experience you have, it's the proportion that is normal that makes good things happen.

Donald Mitchell and Marc Jones

Although a number of key features of the anatomical and physiological organization of the central visual pathways of higher mammals can be remarkably well developed at birth, others achieve their adult status more gradually. Nevertheless, even those characteristics that appear remarkably mature at birth, such as the segregated pattern of ocular dominance observed in the granular layer of the visual cortex (area 17) of monkeys, as well as those that appear less developed, can change in a dramatic fashion in response to abnormal visual input imposed during documented sensitive periods in early postnatal life. The cortical changes induced by such selected early visual exposure can be accompanied by profound and long-standing visual deficits. In this study we demonstrate that with respect to form vision, normal visual input is weighted far more strongly during development than is biased visual exposure such that 2 hours of normal visual exposure each day can outweigh much longer periods of deprivation so as to permit the development of normal vision in both eyes. This result is not what would be expected by a passive process whereby neural connections are continually shaped or refined in response to the integrated visual input during the appropriate sensitive period, but instead suggest a process which effectively responds preferentially to normal input.

T24) Take the HINT. Actually, Scratch That. Don't Take the HINT. Or if You Do Take the HINT, Rotate Your Chair 45 Degrees.

Dennis P. Phillips, B. Vigneault-MacLean, S.E. Boehnke & S.E. Hall

One of the most common problems encountered by persons with or without hearing loss is a poor ability to discriminate speech that occurs against noisy backgrounds. A clinical test (the Hearing In Noise Test, HINT) has been developed to measure the perceptual benefit obtained in speech discrimination achieved by having a 90-degree azimuthal (eccentricity) separation of speech and concurrent noise. The noise is presented from straight ahead, and the speech signal is presented either from the same location, or from 90 degrees into the left or right auditory hemifield. The "release from masking" effected by the 90-degree spatial separation is reported to average about 6-7 dB. Our lab previously showed that the human perceptual system has channels for auditory azimuth that are hemifield in configuration (Boehnke & Phillips, 1999). The HINT does not exploit this knowledge. We'd expect the maximum benefit of a 90-degree separation to occur when speech and noise are presented in different hemifields, and the minimum benefit to occur when the sounds are presented in a single auditory hemifield. The HINT does not employ either of these configurations. In the present study we show, using the HINT stimulus materials, that spatial release from masking can be increased to about 8.6 dB, or reduced to 1.3 dB, simply by locating the speech and noise sources in opposite hemifields or in the same hemifield, respectively, while maintaining the 90 degree azimuthal separation. The HINT configuration produced an intermediate (6.05 dB) release from masking. These data may be useful to clinicians wishing to enhance the effect size of the existing HINT. The data also provide a new corollary of the tuning of human spatial perception mechanisms, which was originally derived from auditory temporal gap detection studies.

T25) **Ovarian Hormones and Acute Restraint Stress Interact to Modulate BDNF Protein Levels in Rat Brain**

Tamara Franklin and Tara Perrot-Sinal

Abnormal levels of brain-derived neurotrophic factor (BDNF) are associated with major depression, a disorder with a higher incidence in women than men. Stress affects BDNF levels in various brain regions and thus, a heightened stress response in females could contribute to the development of depression. As well, ovarian hormones directly affect brain levels of BDNF RNA and protein. The present experiment investigated the interaction between stress and ovarian hormones on BDNF protein levels in the hippocampus (CA1) and paraventricular nucleus of hypothalamus (PVN). Forty-six female Sprague-Dawley rats were ovariectomized; half received one hour of restraint stress while the other half received control handling prior to sacrifice. Stressed and handled groups received a single injection of either estrogen (E; 48 h prior to sacrifice), estrogen and progesterone (EP; P given at 53 h and E given at 48 h prior to sacrifice) or vehicle (OVX). BDNF protein was quantified using an enzyme-linked immunosorbent enzyme assay (ELISA). In the CA1, there was significantly more BDNF in the EP group compared to E and OVX groups. Relative to handled controls, BDNF levels were slightly elevated in OVX-stressed rats but significantly reduced in E-stressed rats. In the PVN there was significantly less BDNF in both E and EP groups compared to OVX rats but there was no effect of stress and no interaction between hormones and stress in this region. Thus, in some stress-sensitive brain regions, the presence of estrogen alters stress-induced changes in BDNF levels. Studies are currently underway to examine the effects of ovarian hormones on BDNF protein in other brain regions involved in stress.

T26) **Effects of restraint stress exposure on hippocampal neurogenesis in rats and mice.**

Megan Bain

Neurogenesis in the dentate gyrus of the hippocampus persists during the adult life of many species, including mice, rats, monkeys, and humans. There are many factors that regulate neurogenesis; however, stress exposure is of particular interest because of its role in human anxiety and depressive disorders. This study investigated the effects of acute restraint stress exposure on hippocampal neurogenesis in rats and mice. Male Sprague-Dawley rats and CD57BL/6 mice were injected with bromodeoxyuridine (BrdU), a marker of neurogenesis, followed by 3 h of restraint stress via confinement to a small container. Using immunohistochemistry, cell proliferation was assessed by measuring the number of BrdU-labeled cells in the dentate gyrus. To determine whether this restraint stress evoked a typical stress response in these species, we also examined the number of c-Fos-labeled cells in the paraventricular nucleus of the hypothalamus (PVN), a brain area known to be activated during stress. The results show that acute restraint stress significantly inhibited hippocampal cell proliferation in rats, as compared to unstressed controls. In contrast, acute restraint stress in mice resulted in increased hippocampal cell proliferation, relative to that seen in unstressed mice. Restraint stress activated the stress response in both species, as indicated by a significant increase in c-Fos immunoreactivity in the PVN. These results suggest that hippocampal neurogenesis in response to stress exposure is differentially regulated in the rat and mouse. Examining the differences between mouse and rat cell proliferation in response to stress may provide useful information on the mechanisms that regulate neurogenesis and a better understanding of the failures of neurogenesis that may underlie various pathological conditions.

T27) Effects of repeated and acute corticosterone on phosphorylated camp response element binding protein in the rat hippocampus.

Sarah Johnson, Andrea Gregus, Alica Davis, Amanda Wintink, Lisa Kalynchuk

Chronically elevated glucocorticoid levels have been linked to stress-related disorders such as depression and physiological changes within the hippocampus including decreased neurogenesis and neuronal remodeling. These physiological changes may result from decreased cAMP and MAP kinase intracellular cascade activity following repeated exposure to high levels of glucocorticoids over a prolonged period of time. The effects of chronic vs. acute increases in glucocorticoids on cAMP cascade proteins in the hippocampus are currently unknown. Accordingly, the current study assessed the effects of chronic vs. acute corticosterone (CORT) exposure on phosphorylated cAMP response element binding protein (pCREB) as a marker for changes in cAMP and MAP kinase intracellular cascades in the male rat hippocampus. Rats received either a single CORT (40mg/kg s.c.) or vehicle injection every day for 21 consecutive days. An additional two groups of rats were handled every day for 20 days and received a single CORT or vehicle injection on day 21. Rats were perfused intracardially 48 hours after the final injection, and hippocampal tissue was immunohistochemically labeled for pCREB. Results are considered in the context of stress and glucocorticoid-based models of depression.

T28) WHAT IF ... you do faint in front of the whole class?? SO WHAT! ... On your wedding day, you will not remember

Denise Bernier, Margo Watt, Sherry Stewart and Cheryl Birch

We conducted a brief cognitive behavioral (CBT) group intervention with first year undergraduate female students. Our objectives were twofold. First, we attempted to replicate the findings of other studies that have examined the efficacy of a brief CBT for panic disorder. We also extended the earlier work by examining, in a non-clinical population, the efficacy of this treatment on lowering levels of anxiety sensitivity, which is a known and established risk factor for the development of anxiety disorders. Secondly, we intended to lay the groundwork for the development of an intervention manual designed for use with groups of high anxiety sensitive university students. Preliminary data will be presented, illustrating the effects of this approach on levels of anxiety sensitivity.

T29) Oxytocin in the Plasma and Cerebrospinal Fluid of Male Rats: Effects of Circadian Time, Light and Stress.

Karthika Devarajan & Benjamin Rusak

Oxytocin (OXT), a neuropeptide known to be involved with reproductive and social behaviours, has also been implicated in responses to stress and depression. Past work has shown that OXT is regulated by circadian phase in the brain, plasma and CSF of monkeys; however, circadian regulation of OXT expression has only been reported in the brain and plasma of the rat. In the present study, we examined the effects of 30 min light pulses at two different circadian phases (mid subjective day and late subjective night) on OXT in circulating plasma, cerebrospinal fluid (CSF) and hypothalamic tissue adjacent to the third ventricle. An enzyme immunoassay was used to measure OXT levels in the plasma and CSF, while immunoreactivity was studied in the hypothalamus. We found that the previously reported circadian difference in plasma OXT levels in the rat disappears when the rats have been previously habituated to handling and injections. Therefore, we speculate that the previously reported OXT rhythm in rat plasma may actually reflect a stress response to experimental procedures expressed selectively at some circadian phases. We also showed for the first time that circadian phase and brief light pulses significantly affect OXT levels in the CSF of the rat.

T30) **Discriminating diode detector demystifies disquieting doubts documenting dyed dextrans' diffusional dribble.**

Steve Shaw, Dalhousie and Ami Fröhlich, MSVU

The insect blood-brain barrier offers a unique model system in which to explore the evolutionary progression of glial function in the CNS. The barrier resides in a superficial layer of glial cells covering the CNS. The powerful barrier in advanced insects like Diptera (flies) evolved from a barrier-less precursor state present in insects' crustacean ancestors, somewhere back around the Carboniferous.

We've been looking for the evolutionary step at which an effective barrier first appeared. A place to start might be the present-day CNS of the alleged Most Primitive Living Insect, a wingless bristletail that inhabits local sea cliffs and looks similar to domestic silverfish. A way to probe the barrier is to inject a tracer into the blood and follow it to see if it gets into the CNS quickly. The tracer of choice has been lanthanum, which can be localized in the electronmicroscope. It seems to be largely excluded from the bristletail's CNS, pointing to the emergence of some sort of a barrier very early on in insects. A major drawback to using lanthanum is that its detection threshold is very high, perhaps 1 part in 10: if it were able to diffuse into the CNS to achieve a concentration up to 10 percent of that in the blood, even this high level could not be detected reliably. We therefore explored whether the detection limit can be lowered significantly by switching to dextrans as light microscope tracers. These inert bacterial polysaccharides are available with fluorescent labels in a range of molecular weights. Our dextrans do appear to be completely excluded by the fly's CNS barrier, gauged by visual observation. We have developed a new approach to quantifying this exclusion, using optical averaging on thin sections of the CNS, measured with a computerized microfluorometer which improves the detection limit markedly. In the fly's CNS, the brightest dextran was excluded from the CNS to better than one part in several thousand, after correcting for the limiting background autofluorescence of the neural tissue. The bristletail's CNS fared much worse when tested with dextrans and appeared quite leaky, somewhere between the extremes of crustaceans and advanced insects. Overall, dextrans emerge as the tracers of choice for probing such barrier systems, where extreme fine-structural localization is not required.

POSTERS

P1) Behavioural and histological analysis of the effects of mup75-saporin in an effort to create a mouse model of Alzheimer's Disease

Lisa M. Currie and Richard E. Brown

A loss of cholinergic cells in the basal forebrain produces memory deficits similar to those in Alzheimer's Disease (AD) (Francis, P.T. et al., 1999, *J. Neurol. Neurosurg. Psych., 66*, 137-47). Following injection of 192-IgG saporin, a neurotoxin targeting p75 cells in the cholinergic basal forebrain, into the medial septum, rats have a deficit in spatial working and reference memory (Shen, J. et al., 1996, *Behav. Neurosci. 110*, 1181-86). The destruction of cholinergic cells in the mouse brain using mup75-saporin (a mouse version of 192 IgG saporin) resulted in a dose-dependant cell loss in the medial septum, and deficits in spatial learning in the Morris Water Maze (Berger-Sweeney, J. et al., 2001, *J. Neuroscience, 21*, 8164-73). Our study extended that of Berger Sweeney et al., (2001) with a more comprehensive analysis of behavioral deficits associated with the injection of mup75-saporin. Using the results of Berger-Sweeney et al., (2001). One of three doses of mup75 saporin (3.6 μg/μl, 1.8 μg/μl, 0.9 μg/μl, or a PBS control) was infused bilaterally into the lateral ventricles of male C57BL/6J mice using a 30 gauge Hamilton syringe. Starting 15-20 days after drug injection, the mice were tested in the Open Field, Elevated Plus Maze, Beam Walking, Rotarod, Morris Water Maze, and a modified win-shift paradigm of the Radial Arm Maze. Following behavioral testing, the mice were inter-cardially perfused using 4% paraformaldehyde in PBS, and the brain tissue was retrieved, and stained for VachT, AchE, and Nissl.

Behavioral analysis indicated no significant effects of mup75-saporin across tasks. However, when body weight was used as a covariate in the Radial Arm Maze, group differences were found on performance measures (latency and total arm entries). Cell counts in the nucleus basalis magnocellularis indicated no significant cell loss due to mup75- saporin. A difference of fiber density in the hippocampus was found between the control and drug groups indicating damage to the septo-hippocampal pathway. Analysis of the superficial layer of Purkinje cells in the cerebellum found no difference across drug groups, indicating that the toxin did not affect the cerebellum. These results indicate that mup75-saporin caused very little cell loss in the nucleus basalis magnocellularis and cerebellum and very few behavioural deficits. It is hypothesized that there may not be as many p75 cells in the mouse brain as in the rat. In addition, other neurotransmitter pathways may be compensating for any cell damage caused by the neurotoxin.

P2) Gradients of attention in visuo-spatial neglect after right hemisphere stroke

Beverly Butler and Gail Eskes

According to Kinsbourne's theory of attention, the two hemispheres of the brain contain opponent, mutually inhibitory vectors to direct attention to different spatial locations; the left hemisphere vector directs attention to the right side of space, while the right hemisphere vector directs attention to both sides of space. Thus, left-sided neglect occurs when the right hemisphere is damaged, resulting in an impaired leftward attentional vector and a disinhibited rightward attentional vector. This theory also suggests that individuals with neglect will show a linear gradient of decreasing attention from right to left. The present study investigated whether individuals with neglect show attentional gradients in both peripersonal space (within arms reach) and extrapersonal space (beyond arms reach). Subjects with right hemisphere damage with or without neglect, and healthy controls searched for visual targets on matched sheets placed in peripersonal and extrapersonal space. A right-to-left gradient of decreasing target detection was seen in both peripersonal and extrapersonal space in the neglect group, but not in the control groups. Double dissociations of gradients in peripersonal and extrapersonal space were also found. These results support Kinsbourne's theory and suggest independent neural mechanisms of attentional control in the right hemisphere related to spatial reference planes.

P3) **Expression pattern of neurotrophic factors and their receptors in the nervous system that develops in the absence of skeletal myogenesis**

Slobodan Mirkovic, Jagdish Gupta, Anne C. Belliveau and Boris Kablar

It is well established that during development motor neurons compete for their target, the skeletal muscle, to avoid programmed cell death. Human disorders characterized with motor control disturbances and human muscular diseases provide some evidence for that. However, it is still largely unclear whether any of the neurotrophins are of unique importance in preventing the death of developing motor neurons. We employed wild-type and *Myf-5-/-:MyoD-/-* embryos (that completely lack skeletal muscle) to test *in vivo* muscle-related neurotrophic requirements of skeletal muscle associated neurons [i.e., MMC (median motor column) and LMC (lateral motor column) motor neurons in the spinal cord and proprioceptive neurons in the dorsal root ganglia (DRG)]. To determine which neurotrophic factor and receptor is important for survival of MMC, LMC and DRG neurons, we assessed the expression pattern of muscle-secreted neurotrophic factors (e.g., BDNF, NT-3, NT-4/5) and their receptors (e.g., Trk B and Trk C) in mutant and control neural tissues by immunohistochemistry. Together, our results indicate that alterations in the expression pattern of neurotrophins and their receptors in the spinal cord and DRG are related to the decreased ability of MMC, LMC and DRG neurons for survival.

P4) **And The Beat Goes On... Life After Heart Surgery**

Lowe-Pearce, C., McGlone, J., Hirsch, G., Legare, J-F., Peters, V.

Controversy exists over recovery from coronary artery bypass grafting (CABG). A healthy control group (n=36) was compared with patients (n=44) to contrast three different Reliable Change Indices (RCI), and then, to contrast outcomes based on incidence (RCI) versus group means (ANCOVA) six weeks prior to surgery and seven months later. Gain, no change, or loss on a composite neuropsychology battery and Quality of Life (SF-36) were identical using each RCI formula. Cognitive change was <u>not</u> significantly different between groups over time, and patients reported significantly improved physical health compared to controls. More patients reported positive <u>and</u> negative mental-health changes than controls, a difference absent from group analyses.

P5) The effect of a single versus repetitive pain experience during infancy on anxiety and spatial learning in juvenile mice.

Lianne Stanford, Matthew Darrah, Carrie Kelly, & Heather Schellinck

Exposure to pain during development has long been hypothesized to disrupt behaviour in later life. (Hebb,1949, Organization of Behaviour). Recent evidence has shown that rats subjected to pain as infants are more anxious as adults (Anand & Scalzo, Bio. Neonate, 77, 2000). In Experiment One, we investigated the influence of early pain experience on anxiety and learning and memory in mice. Neonatal mice were exposed to a one time acute pain, i.e., a tail snip at postnatal day 8 or repetitive pain, i.e., paw pricks on postnatal days 8-14. When tested in an elevated plus maze on Day 30, both acute and chronic pain mice showed behavioural changes indicative of increased anxiety compared with control animals. The effects included a reduction in open arm entries, percent time spent in the open arms and a decrease in head dips. In addition, chronic pain animals showed a greater decrease in time spent in the open arms and in number of entries into the open arms as well as an increase in stretch attends compared with acute pain animals. There were no differences in the number of line crossings in the experimental and control animals. This pattern of results would suggest that the pain exposed animals showed an overall increase in risk assessment, a behaviour associated with hypervigilance in humans. When tested in the Morris Water Maze, these mice demonstrated an intact working memory in a three-day acquisition task as well as in a probe trial and reversal trials. Thus, it appears that their increased anxiety did not adversely affect their performance in a spatial working memory task. In a second experiment, the effect of environmental enrichment on decreasing anxiety in mice experiencing early neonatal pain was assessed. There was no difference in anxiety between control and experimental animals when tested in the elevated plus maze. The implications of this failure to replicate the results of the initial effect are discussed.

P6) Do eyes act like arrows? Responding to nonpredictive directional cues in children and adolescents

Amy MacPherson, Chris Moore, & Chris Kelland Friesen

Covert orienting to socially significant gaze cues and more symbolic arrow cues was recently examined in a manual detection task with preschoolers and adults (Ristic, Friesen, & Kingstone, in press). Significant cueing effects were observed in both conditions for both age groups, contrary to the hypothesis that for young children arrows are less well-learned than they are for adults and would therefore make less effective cues than gaze cues (which are believed to be processed reflexively). The present cross-sectional study used the same stimuli and design as Ristic et al., but with a localization task, to examine the range of ages between those tested in their study. One hundred and ten 5- to 17-year-olds were recruited and tested at a summer camp. They participated in two counterbalanced cueing conditions, making keypress responses to peripheral targets appearing, at varying cue-to-target stimulus onset asynchronies (SOAs), to the left or right of centrally-positioned nonpredictive directional cues. In the Gaze Condition, the direction of gaze of a schematic happy face served as the cue, and in the Arrow Condition the direction of an arrow served as the cue. Participants made a speeded response by pressing either a left or right computer key to indicate target location. Each block of 52 randomly ordered trials included two each of 24 trial types representing the factorial combination of cue validity (valid or invalid), target location (left or right), target identity (cat or snowman) and SOA (195, 600, or 1005 ms), along with 4 catch trials. The 5- to 10-year-olds (N = 49, M = 8y4m) completed one block in each condition, while the 11- to 17-year-olds (N = 61, M = 14y) completed two blocks in a row in each condition. As expected, we found that participants in both age groups responded more quickly to targets appearing on the side of the screen indicated by the cue. Although both reaction time (RT) and the magnitude of the RT advantage decreased as age increased, the overall pattern of faster responding to targets on the cued side was consistent across age groups in both the Gaze and Arrow Conditions. The present study provides further evidence that when gaze and arrow cues are nonpredictive they can produce similar effects, and it adds the new finding that the cueing effects triggered by nonpredictive centrally-presented cues remains consistent from the preschool years through to adulthood.

P7) **Eye movements are not required for fast manual responses to visual target perturbations.**

David A. Westwood (Dalhousie University) & James Danckert (University of Waterloo).

Previous studies have shown that reaching movements can be quickly updated when the position of the target is unexpectedly changed during the course of the response (i.e., the double-step reaching paradigm). Because the change in target position typically takes place during a saccadic eye movement, perceptual awareness of the perturbation is suppressed (i.e., saccadic suppression). Nevertheless, the eyes move quickly (and unconsciously) to fixate the new target position before the reaching movement is complete. Therefore it is unclear whether the updating of the reaching movement is driven by a visual (i.e., the position of the target on the retina) or non-visual signal (i.e., gaze direction). We sought to distinguish between these possibilities by comparing reaching performance in a double-step task with free gaze (i.e., subjects were free to saccade to the target when it appeared) and central fixation (i.e., subjects maintained central fixation during the course of the trial). If reaching corrections are driven by a non-visual signal related to eye position then performance should be impaired in the central fixation condition (since eye position and target position are decoupled). In contrast to this prediction, our results show that performance is equivalent in the two conditions; reaching adjustments were as fast, accurate, and frequent in the free gaze and central fixation conditions. Although knowledge of eye position is undoubtedly important for the control of reaching movements, the reaching control system has fast access to visual information about the position of targets on the retina; this information can be used to implement fast and accurate adjustments to ongoing reaching movements. In other words, the control of reaching movements is not a simple case of reaching to where the eyes are looking.

P8) **Periodic stimulation (of the vagal nerve for treatment of intractable seizures) may be less than it is cracked up to be.**

McGlone, J.[1], Clarke, D.[2], Sadler, R.M.[3], Valdivia, I.[1], and Penner, M.[1] Departments of Psychology[1], Surgery[2] and Medicine (Neurology)[3]

In an experimental paradigm, Clarke et al. (2000) found that verbal recognition memory was enhanced immediately after stimulation of the vagal nerve in humans with epilepsy. If this effect were more permanent, the prediction could be made that patients receiving vagal nerve stimulators might experience enhancement of memory functioning on standardized clinical tests. Three groups of patients with epilepsy underwent a broad neuropsychology evaluation including a memory assessment (WMS-III) and self-rating of memory functioning on two occasions, i.e., before and 12 months after implantation of a vagal nerve stimulator (n = 16), or a temporal lobectomy (n = 10), and a medication only control group (n = 9) seen 12 months apart. Depression scores (Geriatric Depression Scale), seizure frequency, and Quality of Life (QOLIE-89) were also analyzed. Grouped data were subjected to an Analysis of Variance and individual change scores were subjected to a reliable change index analysis. **Results:** The patients with vagal nerve stimulators were not superior to the other groups on any outcome measure one year after treatment. Similarly, the percentage of patients who improved beyond measurement error did not differ in the three patient groups. **Conclusions:** Vagal nerve stimulation did not reduce seizure frequency or enhance memory functioning or quality of life in a clinical sample of patients with medically intractable epilepsy after 12 months of treatment. Our negative findings based on matched samples contrasts with the positive outcomes published in industry funded, but less well controlled studies. The results highlight the need for non-industry funded granting agencies to embark on multi-centre research on this new technology.

P9) **Abnormal Development of the Diaphragm in *mdx:MyoD-/-^{9th}* embryos leads to pulmonary hypoplasia**

Mohammad R. Inanlou and Boris Kablar

Introduction: *In vitro* studies have shown that mechanical factors play an important role in cell cycle kinetics and cell differentiation of the lung through an unknown mechanochemical signal transduction pathway. In the current study we evaluated the *in vivo* role of mechanical factors due to fetal breathing movements (primarily executed by the diaphragm) on lung growth and development by using genetically engineered mouse embryos.
Materials and Methods: Lung growth and development of *mdx* and the ninth backcrossed generation of *mdx:MyoD-/-* embryos (designated as: *mdx:MyoD-/-^{9th}*) were compared to that of the wild-type littermates (and to each other) at the embryonic day 18.5 using histological studies and immunohistochemistry.
Results: The mesenchymal compartment of the diaphragm was intact and no herniations were observed in the embryos. By contrast, the skeletal muscle compartment of the diaphragm was significantly reduced in *mdx:MyoD-/-^{9th}* embryos. No abnormalities in lung organogenesis were observed in *mdx* term embryos, whereas lung hypoplasia was detected in *mdx:MyoD-/-^{9th}* embryos. In the hypoplastic lung, the number of proliferating lung cells was lower in comparison to the wild-type and *mdx* littermates, while the thyroid transcription factor-1(TTF-1) gradient

was not maintained. Surprisingly, no difference was observed in the apoptotic lung cells distribution and occurrence in *mdx:MyoD-/-*^{9th} embryos.

Conclusions: Together, it appears that mechanical forces generated by contractile activity of the diaphragm muscle play an important role in normal lung growth and development by affecting cell proliferation and TTF-1 expression.

P10) Potentially protective effects of Ritalin

Martin Williamson and Richard E. Brown

Methylphenidate (MPH, Ritalin), a central nervous system stimulant, is widely used for the treatment of Attention-Deficit Hyperactivity Disorder (ADHD), a chronic childhood disorder with a prevalence of 3 to 6% in school-aged children. In recent years, the therapeutic use of MPH for the management of ADHD has increased substantially in children and also in adults. Furthermore, as MPH is increasingly prescribed to adult women to treat ADHD, there is an increasing danger of taking MPH during pregnancy. Little is known about the effects of MPH during pregnancy, therefore in this study, pregnant C57BL/6J mice were given daily sub cutaneous injections of MPH (5 mg/kg) or saline during embryonic day 4 to 18. Since postnatal environmental enrichment can exert profound and positive neurobehavioural changes in animals, pups were reared in an enriched or isolated environment. On postnatal day 60, the pups underwent behavioural testing in the elevated plus maze, open field, rotarod and Morris water maze. Prenatal MPH exposure caused lasting changes in behaviour, illustrated by decreases in emotionality and increases in exploration and locomotion, while having no effect on motor and spatial learning performance. Pups reared in an enriched environment showed similar reductions in emotionality, and increased exploratory and locomotor behaviour, combined with improved spatial learning performance. Of particular interest is the finding that the effect of prenatal MPH injections on behaviour is similar to that of environmental enrichment. Future studies will focus on the mechanisms mediating the behavioural alterations and the potential stimulating or protective role of enrichment and stimulant drugs such as MPH.

BALLOT

Most Unusual Abstract Title

Title ____

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Author ____